



High-Speed Digital PCB Design

When wires become transmission lines

DDR • USB • HDMI • PCIe • Ethernet routing rules

Audience: Intermediate Engineering Students

Impedance • Length matching • Diff pairs • Stackup • Vias



What is 'High Speed'?

It's not the clock — it's the edge

- ▶ Rise-time matters more than clock frequency
- ▶ Rule of thumb: trace length > rise-time × 0.5 ns/cm → transmission line behaviour
- ▶ 1 ns edge contains energy up to ~350 MHz
- ▶ Modern logic: 100-500 ps edges — almost any trace is high speed
- ▶ Once a trace IS a transmission line, impedance, reflections, and crosstalk all matter
- ▶ Common 'high-speed' interfaces: USB 2.0/3.0, HDMI, DDR, PCIe, Gigabit Ethernet
- ▶ Even slow signals (UART, SPI) can need care at long board lengths

Characteristic Impedance

The single most important high-speed parameter

- ▶ $Z_0 = \sqrt{L/C}$ per unit length of trace
- ▶ Depends on: trace width, dielectric thickness, dielectric constant
- ▶ 50 Ω is the de-facto standard for single-ended
- ▶ Differential pairs: 90, 100, 120 Ω depending on standard
- ▶ Stackup determines impedance — controlled-impedance PCB
- ▶ Tools: KiCad calculator, Saturn PCB toolkit, Polar SI9000

Common targets

USB 2.0/3.0 differential — 90 Ω
HDMI — 100 Ω
DDR3/4 — 40-50 Ω single-ended
Ethernet (RJ45) — 100 Ω diff
PCIe — 85 Ω diff

How to get there

1. Pick stackup with vendor (e.g. JLCPCB JLC04161H)
2. Use vendor-supplied impedance numbers
3. Annotate traces as controlled-impedance
4. Vendor adjusts width as needed

Stackup Design for High Speed

Layer order matters as much as count

- ▶ Minimum 4 layers for any high-speed signal
- ▶ Layer 1: signals • Layer 2: GND plane • Layer 3: PWR • Layer 4: signals
- ▶ Reference plane (GND) must be CONTINUOUS under every high-speed trace
- ▶ Differential pairs route on signal layers adjacent to GND
- ▶ 6-layer adds an extra GND for better return paths and inner-layer routing
- ▶ DDR4 / PCIe often need 8-10 layers for proper plane allocation
- ▶ Choose dielectric (Dk) and thickness with the fab early — drives impedance numbers

Differential Pair Routing

Tightly coupled, length-matched, impedance-controlled

Tight coupling

Spacing typically equals trace width.
Keeps fields coupled to each other.

Length match

Match within 5 mil (0.13 mm).
Add serpentine 'wiggles' as needed.

Symmetric routing

Same number of vias, same path lengths,
same layer transitions.

Avoid splits

Never cross a plane split or gap —
ground return discontinuity.

Maintain impedance

Width changes break Z_0 .
Keep constant from driver to receiver.

Bend together

Both traces follow the same path.
Keep skew low.

Length Matching

Why bit skew kills timing

- ▶ Each trace has a propagation delay: ~150 ps/inch in FR-4
- ▶ Skew = difference in arrival time between matched signals
- ▶ DDR4 1600 MT/s — UI = 625 ps; skew budget < 50 ps
- ▶ USB 3.0 differential — pair-skew < 15 ps
- ▶ Length match by adding serpentine (snake) traces
- ▶ Match in groups: byte lane (DDR), pair (USB), or bus

DDR3/4 rules

1. Match within byte lane to <30 ps
2. Match byte-lane to clock <50 ps
3. Strobe DQS to DQ <20 ps
4. Use vendor-recommended topology (fly-by / T-branch)

How to length match

KiCad: select traces → "Tune length" → enter target length.
Altium / Allegro have similar tools.
Never guess — use the tool's report.

Vias — The Hidden Discontinuity

Every via is a small impedance bump

- ▶ A through-via at 1.6 mm board ~ 1 nH inductance + ~ 0.3 pF capacitance
- ▶ Each via causes a small impedance discontinuity $\rightarrow$ reflection
- ▶ Minimise via count on high-speed signals
- ▶ Add return-path vias (ground stitching vias) next to signal vias
- ▶ Microvias / blind vias for very-high-speed (USB 3.0+, PCIe 4.0+)
- ▶ Back-drill (controlled-depth drill) removes via stubs in high-frequency designs
- ▶ Anti-pad sizing affects impedance — work with the fab

Crosstalk

When neighbouring traces inject noise

- ▶ Coupling between adjacent traces — capacitive + inductive
- ▶ Falls quickly with trace spacing
- ▶ 3W rule: spacing $\geq 3\times$ trace width on signal layer
- ▶ Worse for parallel-routed long traces
- ▶ Mitigation: increase spacing, add guard trace, route on different layers
- ▶ Differential pairs: less coupled to outsiders, more coupled within pair

3W rule

Keep spacing between non-related traces $\geq 3\times$ trace width. This drops crosstalk by $\sim 10\times$.

Guard traces

A grounded trace between two signals further isolates them. Stitch the guard to GND every $\lambda/20$.

Return-Path Discipline

Current always flows in loops — at high speed, paths are below the trace

- ▶ Every signal trace has a return current
- ▶ At high frequency, return current flows DIRECTLY beneath the trace
- ▶ Splits, gaps, or missing copper force return to detour — bad EMI
- ▶ When changing layers, also change reference plane → add a stitching via
- ▶ Best layer transitions: signal → GND → signal (no plane change needed)
- ▶ Sub-optimal: signal → PWR → signal (return must hop through caps)
- ▶ Always provide a continuous reference plane for high-speed signals

USB 2.0 & 3.0 — Practical Rules

Most common high-speed interface in student projects

Parameter	USB 2.0	USB 3.0 (SS)
Data rate	480 Mbps	5 Gbps
Differential pair	D+/D− 90 Ω	SSTX/SSRX 90 Ω each
Length match	< 50 mil pair-skew	< 5 mil pair-skew
Max trace length	< 6 inches typical	< 4 inches typical
Layer count	2 (4 better)	4+ required
Reference plane	GND, continuous	GND, continuous, multiple stitching vias
ESD protection	TVS array on data lines	Differential TVS, low capacitance
Common parts	USBLC6 (TVS)	ESD7104, TPD4S012

Connector matters

Gigabit Ethernet

MAC ↔ PHY ↔ Magnetics ↔ Cable

- ▶ 100 Ω differential, 4 pairs (MDI[0..3])
- ▶ Magnetics (transformer + common-mode choke) between PHY and RJ45
- ▶ Center tap of each pair on the PHY side goes to PHY-supplied bias
- ▶ Cable side center taps go to chassis through Bob Smith termination (75 Ω + cap)
- ▶ Length-match all 4 pairs within ~50 mil of each other
- ▶ RJ45 connectors with integrated magnetics simplify layout
- ▶ Critical: ESD protection on all 8 data lines (cable can ESD)

DDR3 / DDR4 Routing

The hardest commonly-seen high-speed interface

- ▶ Match in byte-lanes: DQ[0..7] + DQS + DM grouped together
- ▶ Topology: fly-by for address/command, T-branch alternatives
- ▶ Single-ended impedance $\sim 40\text{-}50\ \Omega$ (DDR4); diff strobe $\sim 80\text{-}100\ \Omega$
- ▶ On-die termination (ODT) reduces external resistor count
- ▶ Skew between data and strobe $< 20\text{ ps}$ for DDR4
- ▶ VTT termination on the far end of the bus
- ▶ Reference voltage VREF must be clean — separate LDO
- ▶ Layer count usually 6-10+ for DDR4 dense designs

Power Integrity (PI) for High-Speed

Switching currents demand local energy

- ▶ Every IC needs decoupling caps at every power pin
- ▶ Place caps ≤ 1 mm from the pin — short trace + via to plane
- ▶ Use multiple values in parallel: 100 nF + 10 nF + 1 nF cover different frequencies
- ▶ Bulk cap (10 μ F) per power region
- ▶ Power plane (not trace!) for any IC drawing > 100 mA dynamic
- ▶ PDN (Power Distribution Network) impedance targets — simulator helps
- ▶ Stitch power & GND planes with vias every ~ 5 mm

Termination Techniques

Match impedance at one or both ends

Series source

$R = Z_0 - Z_{\text{driver}}$, at the driver.
Simple, used in CMOS clock fanout.

Parallel end

$R = Z_0$ to ground at receiver.
DC power waste, but matches well.

Thevenin

Split resistors to power & ground.
Biased, used on backplanes.

AC parallel

$R + C$ in series to GND at end.
No DC current; good for slow buses.

Differential

100 Ω across diff pair at receiver.
Standard for USB/HDMI/Ethernet.

ODT

On-die termination in modern IC.
No external R needed.

Simulation & Verification

Tools to verify before you build

- ▶ IBIS models from IC vendors describe driver / receiver behaviour
- ▶ SI simulators: HyperLynx, Sigrity, ADS, Polar Si8000
- ▶ Run pre-layout simulation to check topology before drawing PCB
- ▶ Post-layout simulation extracts PCB parasitics; verifies eye diagrams
- ▶ TDR (Time Domain Reflectometry) measures impedance on real boards
- ▶ VNA (Vector Network Analyser) measures S-parameters of links
- ▶ KiCad's built-in simulator is limited — use specialist tools for high-speed

Common High-Speed Mistakes

Caught by post-build EMI testing or worse — silent data corruption

Crossing plane gaps

Return path detours →
EMI, jitter, signal degrades.

Long via stubs

Resonates near band-edge.
Use back-drill or microvia.

Skew over budget

Bit errors that come and go
with temperature.

Wrong impedance

Reflections create ringing.
Receiver mistimes data.

Insufficient decoupling

Switching noise contaminates
clock & data.

BGA escape too dense

Internal layers can't fan out.
Need via-in-pad / HDI.

Key Takeaways

Key takeaways from this lecture

Edge rate matters

Not the clock — the edge defines high-speed.

Controlled-Z everywhere

Stackup + vendor partnership.

Return paths sacred

Never cross a plane split.

Length match groups

Within byte-lane / pair / bus.

Decouple aggressively

Multi-value caps near every IC pin.

Simulate before fab

IBIS + SI tools save weeks.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ Eric Bogatin — 'Signal & Power Integrity Simplified' (definitive reference)
- ▶ Howard Johnson — 'High-Speed Digital Design' (classic black-magic book)
- ▶ Stephen Hall — 'Advanced Signal Integrity for High-Speed Digital Designs'
- ▶ Saturn PCB Toolkit — free impedance calculator
- ▶ Polar Instruments — controlled-impedance reference tables
- ▶ Robert Feranec (YouTube) — DDR, USB, HDMI walkthroughs
- ▶ Intel + AMD design guides — exact stackup recipes
- ▶ JEDEC, USB-IF, HDMI Forum, PCI-SIG — standards documents

Thank you • Build something this week.